

Question:

Raymond and Samantha are cousins. Raymond was born 6 years before Samantha. Raymond had a son at the age of 23. If Samantha is now 31, how many years ago was Raymond's son born?

Reasoning + Answer:

We are given :

- Raymond was born 6 years before Samantha .
- Raymond had a son at the age of 23 .
- Samantha is now 31 .
- We are to find : ****How many years ago was Raymond's son born ?****

Step 1 : Determine Raymond's age when he had the son

Raymond had a son at the age of 23 .

Step 2 : Find out when Samantha was born

Samantha is now 31 .

Raymond was born 6 years before Samantha , so :

- If Samantha is 31 now , Raymond was born 6 years earlier than Samantha .

So Raymond was born at :
Samantha's birth age minus 6 years = $31 - 6 =$ ****25 years ago****?

Wait ⚠ let's clarify .

Let's define the present time :

Let ⚠s suppose the current year is now .

- Samantha is 31 ⚠ she was born 31 years ago .
- Raymond was born 6 years before Samantha ⚠ so Raymond was born at $31 - 6 = 25$ years ago .

So :

- Raymond was born 25 years ago .
- He had a son at age 23 ⚠ so son was born when Raymond was 23 .

Raymond was 23 at :
 $\text{Birth year of Raymond} + 23 \text{ years} \hat{=} 25 \text{ years ago} + 23 \text{ years} =$ ****2 years ago****

So Raymond's son was born ****2 years ago**** .

⚠ Therefore , the son was born 2 years ago .

Answer : ****2 years ago****

Let ⚠s write it formally :

- Samantha is 31 ⚠ born 31 years ago .
- Raymond was born 6 years before Samantha ⚠ born at $31 - 6 = 25$ years ago .
- Raymond had a son at age 23 ⚠ 23 years after his birth ⚠ $25 + 23 = 48$ years from now ? Wait ⚠ no .

Wait ⚠ we need to be careful with timeline .

Let ⚠s define timeline in terms of years from now .

Better approach : set the current time as year 0 .

Let time now be ****0**** .

- Samantha ⚠s age now : 31 ⚠ she was born at time = -31
- Raymond was born 6 years before Samantha ⚠ so Raymond was born at time = $-31 - 6 = -37$

Raymond had a son when he was 23 ⚠ which was at time :
 $-37 + 23 =$ **** -14 ****

So the son was born at time -14 ⚠ which is ****14 years ago****? Wait .

Wait : -14 means 14 years ago ? Yes .

But let's double-check .

- Samantha is 31 ⚠ born at time -31
- Raymond was born 6 years before Samantha ⚠ born at -37
- When Raymond was 23 , he had a son : so son born at $-37 + 23 = -14$ ⚠ ****14 years ago****

So the son was born ****14 years ago**** .

Wait ⚠ that contradicts the previous .

Ah ! Where was the mistake ?

"Raymond was born 6 years before Samantha " ⚠ so if Samantha is 31 now , then :

Samantha born at : year = -31

Raymond was born 6 years earlier ⚠ born at : $-31 - 6 = -37$

Then , Raymond had a son at age 23 ⚠ so son born at : $-37 + 23 =$ **** -14 ****

So son born 14 years ago .

Yes !

But earlier , I said Raymond born at 25 years ago ⚠ that's wrong .

Samantha is 31 ⚠ born 31 years ago .

Raymond born 6 years earlier ⚠ so $31 + 6 = 37$ years ago